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Inventor(s)	Kelpie; Patricia et al.

Apparatus and method for education and learning

Abstract

Embodiments in accordance with the present disclosure provide an apparatus that facilitates education and learning through tactile sensory demonstration and action. The apparatus includes a first part and a second part attached to the first part. The first part may be worn by a guide, while the second part may be worn by a child or a student undergoing training. The apparatus allows the guide to control the movements of the child or the student.

Inventors: **Kelpie; Patricia (Londonderry, GB), Munoz-Condis; Maria (Dumont, NJ)**

Applicant: **STAR 3 GROUP, LTD.** (Londonderry, GB)

Family ID: **64267724**

Assignee: **STAR 3 GROUP, LTD. (Londonderry, GB)**

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Primary Examiner: Hull; James B

Attorney, Agent or Firm: Hard IP LLC

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation application and claims priority to U.S. patent application Ser. No. 16/182,281 filed on Nov. 6, 2018, which claims the benefit of both U.S. Provisional Application Ser. Nos. 62/582,883 filed on

Nov. 7, 2017, and 62/583,220 filed on Nov. 8, 2017, the entire contents of which are hereby incorporated by reference in their entirety.

FIELD OF THE INVENTION

(1) Embodiments in accordance with the present disclosure relate to an apparatus and a method that facilitates education and learning. In particular, embodiments in accordance with the present disclosure relate to a hand-over-hand apparatus and a method for facilitating education and learning of a physical task through tactile sensory demonstration and action.

BACKGROUND

(2) Autism is one of the most common and prevalent neurodevelopmental disorders. Autism typically occurs on a spectrum of severity and is characterized by deficits in communication and social skills, and the presence of rigid, repetitive behaviors. People having autism may require lifelong care.

(3) A parent or a caregiver of a child with autism may also suffer from fatigue and stress due to an increased amount of childcare and continued dependency of the child in performing basic tasks.

(4) At present, there is no known cure for autism and the exact cause is still being investigated. Further, many children and adults on the autism spectrum and/or other developmental disabilities do not have the attention, social skills, and/or learning capabilities to adhere to traditional styles of teaching.

(5) Related arts include various tools or devices to facilitate teaching of basic tasks to students/children with developmental disabilities. For example, a related art provides training scissors that feature a V-shaped design of the finger holes making it easy and comfortable for a teacher's hand to guide the students.

(6) Related art further provides a spandex glove that provides just enough proprioceptive input and compression to facilitate handwriting. Other conventional training devices include shoe lace training kits, dressing frames, pencil grips, and so forth.

(7) However, such devices or tools may have a complicated design and are limited to a particular task. Moreover, such devices or tools may not provide adequate control to a teacher or guide.

(8) Moreover, children with Autism Spectrum Disorder (ASD) may have a sensitivity to certain materials and labels. Specifically, clothes with washing labels, itchy materials and seams may cause sensory discomfort to children with ASD.

SUMMARY

(9) Embodiments of the present disclosure generally relate to a hand-over-hand apparatus or device that facilitates education and learning through tactile sensory demonstration and action. The apparatus provides enhanced teaching experience while teaching basic skills, such as self-care tasks, handwriting, scissor skills, hand motor coordination and so forth, to a disabled or an autistic child. The apparatus also may be helpful to teach skills to persons with certain disabilities such as blindness or deafness, and to persons rehabilitating from physical injuries or diseases that affect motor skills.

(10) Embodiments in accordance with the present disclosure provide an apparatus including two gloves attached to one another such that the apparatus can accommodate two person's hands in a hand-over-hand configuration. The gloves may be made of a fabric-like material. The material may include certain textures, durability and varying properties, such as waterproof or water resistant and easy cleanability. The glove material is not necessarily selected on the basis of its heat insulation properties.

(11) Embodiments in accordance with the present disclosure further provide an apparatus that facilitates education and learning through tactile sensory demonstration and action. The apparatus may be made from a material which is attuned to the sensory needs of children with autism spectrum disorder. Further, the material may include 100 percent soft cotton, with no labels. The apparatus may also comply with all the safety specifications of autism.

(12) Embodiments in accordance with the present disclosure further provide an apparatus that facilitates education and learning through tactile sensory demonstration and action. The apparatus may be available in different colors to make the use of the apparatus fun for children. Further, the apparatus may be available in different sizes (i.e., small, medium and large) to accommodate different sizes of hands. Moreover, the apparatus may provide different range of motion for effective training.

(13) Embodiments in accordance with the present disclosure further provide an apparatus that facilitates education and learning through tactile sensory demonstration and action. The apparatus may serve as a memory building tool which is interactive, easy to use, handy, portable, affordable and lightweight.

(14) Embodiments in accordance with the present disclosure further provide an apparatus that facilitates education and learning through tactile sensory demonstration and action. The apparatus includes a first part and a second part attached to the first part. The first part may be a first fingerless glove, while the second part may be a second fingerless glove. Further, the first glove may be attached on a top side of the second glove via any attachment methods, such as but not limited to, sewing, heat sealing, Velcro® hook and loop fasteners, and so forth. The apparatus may provide greater flexibility and movement due to minimum attachment between the first glove and the second glove.

(15) Embodiments in accordance with the present disclosure are directed to an apparatus that facilitates education and learning through tactile sensory demonstration and action. The apparatus includes a glove configured for receiving two hands. The glove may be a short fingerless glove. The glove may include a top portion and a bottom portion sufficiently spaced apart from the top portion, to allow receiving of two hands simultaneously.

(16) Embodiments in accordance with the present disclosure are further directed to a handover-hand apparatus including a glove. The glove including five cavities, one for each of the four fingers and a thumb. The glove also includes a material banding or band under each of the five cavities. The material of the band or banding may be an elastic material and/or a fabric material. The glove is configured to receive a hand of a person, while the material bandings are configured to receive fingers of another person.

(17) Embodiments in accordance with the present disclosure are further directed to a handover-hand apparatus. The hand-over-hand apparatus includes a first glove and a second glove. The first glove may be attached to the second glove from all the sides and around each finger of the second glove.

(18) Yet other embodiments of the present disclosure are directed to a hand-over-hand apparatus. The hand-over-hand apparatus includes a first glove and a second glove. The first glove may be attached to the second glove at finger tips to allow maximum movement and flexibility. Further, the first glove may be attached to the second glove by any suitable attachment methods, such as sewing, heat seal, Velcro® and so forth.

(19) Other embodiments of the present disclosure are directed to a hand-over-hand apparatus. The hand-over-hand apparatus includes a first glove and a second glove. The first glove may be attached to the second glove at the sides and at the corresponding finger elements to allow greater control and maximum pressure. Therefore, the apparatus may be used for training with sharp objects, for example, a pair of scissors.

(20) Embodiments of the present disclosure also provide a method of training or teaching various tasks to a child or a student suffering from a developmental disability, such as autism. The method uses a hand-over-hand apparatus or tool that facilitates training or teaching through tactile sensory demonstration and action.

(21) Embodiments of the present disclosure further provide a hand-over-hand apparatus or device that has a low production cost per unit, has a low weight, has easy and low-cost packaging and shipping, and has an affordable purchase price so that it is accessible to parents of learning disabled

or autistic children.
(22) Embodiments of the present disclosure also provide a hand-over-hand apparatus or tool that provides an enhanced experience and can shorten a duration required for teaching various skills to children with developmental disabilities and learning challenges. Such skills include self-care tasks, handwriting (pencil grip, letter formation etc.), scissors skills, hand motor coordination, and so forth. Detailed instructions may be provided with the hand-over-hand apparatus to help a teacher or a guide.
(23) These and other advantages will be apparent from the present application of the embodiments described herein.
(24) The preceding is a simplified summary to provide an understanding of some embodiments of the present disclosure. This summary is neither an extensive nor exhaustive overview of the present disclosure and its various embodiments. The summary presents selected concepts of the embodiments of the present disclosure in a simplified form as an introduction to the more detailed description presented below. As will be appreciated, other embodiments of the present disclosure are possible utilizing, alone or in combination, one or more of the features set forth above or described in detail below.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) The accompanying drawings, which are incorporated in and constitute a part of this specification, illustrate several embodiments of the disclosure and, together with the description, explain the principles of the disclosure.
(2) FIG. **1**A illustrates a bottom perspective view of a hand-over-hand apparatus with an adult hand inserted therein, in accordance with an embodiment of the present disclosure;
(3) FIG. **1**B illustrates a bottom perspective view of the hand-over-hand apparatus of FIG. **1**A;
(4) FIG. **1**C illustrates a top view of the hand-over-hand apparatus of FIG. **1**A in use, with one of two hands visible;
(5) FIG. **2**A illustrates a bottom view of a hand-over-hand apparatus in use, with one hand inserted therein, in accordance with an embodiment of the present disclosure;
(6) FIG. **2**B illustrates a top view of the hand-over-hand apparatus of FIG. **2**A, with one hand inserted therein;
(7) FIG. **3**A illustrates a bottom perspective view of a hand-over-hand apparatus, in accordance with an embodiment of the present disclosure;
(8) FIG. **3**B illustrates a side perspective view of the hand-over-hand apparatus of FIG. **3**A in use;
(9) FIG. **3**C illustrates a bottom perspective view of the hand-over-hand apparatus of FIG. **3**A in use;
(10) FIG. **3**D illustrates a top view of the hand-over-hand apparatus of FIG. **3**A;
(11) FIG. **4**A illustrates a top perspective view of a hand-over-hand apparatus, in accordance with an embodiment of the present disclosure;
(12) FIG. **4**B illustrates a side perspective view of the hand-over-hand apparatus of FIG. **4**A in use;
(13) FIG. **4**C illustrates another side perspective view of the hand-over-hand apparatus of FIG. **4**A in use;
(14) FIG. **4**D illustrates a top view of the hand-over-hand apparatus of FIG. **4**A in use;
(15) FIG. **5**A illustrates a bottom perspective view of a hand-over-hand apparatus, in accordance with an embodiment of the present disclosure;
(16) FIG. **5**B illustrates a bottom perspective view of the hand-over-hand apparatus of FIG. **5**A, with one of two hands visible;
(17) FIG. **5**C illustrates a top view of the hand-over-hand apparatus of FIG. **5**A;

(18) FIG. **6**A illustrates a side perspective view of a hand-over-hand apparatus in use, in accordance with an embodiment of the present disclosure;
(19) FIG. **6**B illustrates a bottom perspective view of the hand-over-hand apparatus of FIG. **6**A in use;
(20) FIG. **6**C illustrates a top view of the hand-over-hand apparatus of FIG. **6**A;
(21) FIGS. **7**A and **7**B illustrate various views of a hand-over-hand apparatus, in accordance with an embodiment of the present disclosure;
(22) FIG. **8** illustrates a top perspective view of a hand-over-hand device, in accordance with an embodiment of the present disclosure;
(23) FIGS. **9**A, **9**B, **9**C, **9**D, **9**E and **9**F illustrate various views of an apparatus suitable for use by a younger or smaller patient;
(24) FIGS. **9**G, **9**H and **9**I illustrate usage of the apparatus illustrated in FIGS. **9**A through **9**F;
(25) FIGS. **10**A, **10**B, **10**C, **10**D, **10**E and **10**F illustrate various views of an apparatus suitable for use by a relatively larger patient;
(26) FIG. **10**G illustrates usage of the apparatus illustrated in FIGS. **10**A through **10**F; and
(27) FIG. **11** illustrates a flowchart of a method of teaching a task using a hand-over-hand apparatus, in accordance with an embodiment of the present disclosure.
(28) While embodiments of the present disclosure are described herein by way of example using several illustrative drawings, those skilled in the art will recognize the present disclosure is not limited to the embodiments or drawings described. It should be understood the drawings and the detailed description thereto are not intended to limit the present disclosure to the form disclosed, but to the contrary, the present disclosure is to cover all modification, equivalents and alternatives falling within the spirit and scope of embodiments of the present disclosure as defined by the appended claims.
(29) The headings used herein are for organizational purposes only and are not meant to be used to limit the scope of the description or the claims. As used throughout this application, the word “may” is used in a permissive sense (i.e., meaning having the potential to), rather than the mandatory sense (i.e., meaning must). Similarly, the words “include”, “including”, and “includes” mean including but not limited to. To facilitate understanding, like reference numerals have been used, where possible, to designate like elements common to the figures.

DETAILED DESCRIPTION

(30) The phrases “at least one”, “one or more”, and “and/or” are open-ended expressions that are both conjunctive and disjunctive in operation. For example, each of the expressions “at least one of A, B and C”, “at least one of A, B, or C”, “one or more of A, B, and C”, “one or more of A, B, or C” and “A, B, and/or C” means A alone, B alone, C alone, A and B together, A and C together, B and C together, or A, B and C together.
(31) The term “a” or “an” entity refers to one or more of that entity. As such, the terms “a” (or “an”), “one or more” and “at least one” can be used interchangeably herein. It is also to be noted that the terms “comprising”, “including”, and “having” can be used interchangeably.
(32) The present disclosure is related to a hand-over-hand apparatus or device, and a method of using the hand-over-hand apparatus. The hand-over-hand apparatus includes a first part and a second part attached to the first part. The first part may include any one of a short fingerless glove, a long fingerless glove, a full glove or elastic bandings/bands. The first part may be configured to receive a hand of a first person. The second part may include any one of a short fingerless glove, a long fingerless glove, a full glove or material bands. The material of the material bands maybe, for example, an elastic material and/or a fabric material. Material bands for the first part are usable with a glove for the second part, and material bands for the second part are usable with a glove for the first part. The second part may be configured to receive a hand of a second person. A fingerless glove of the present disclosure may relate to a glove that allows fingers to extend out of corresponding finger elements or projections, i.e., the fingers are not completely enclosed by the

glove. In some embodiments, the first part and the second part may be attached to each other via any attachment methods, such as but not limited to, sewing, hot seal, Velcro® hook and loop fasteners, and so forth. In some other embodiments, the first part and the second part may be integrally formed and may together define a cavity to receive hands of two persons. In an exemplary embodiment, the first part is configured to receive a hand of an adult or a guide, while the second part is configured to receive a hand of a child or a student.

(33) FIG. **1**A illustrates a bottom perspective view of a hand-over-hand apparatus or device **100** (hereinafter referred to as "the apparatus **100**") in use, in accordance with an embodiment of the present disclosure. FIG. **1**B illustrates a bottom perspective view of the apparatus **100**. FIG. **1**C illustrates a top view of the apparatus **100**. Referring to FIGS. **1**A-**1**C, the apparatus **100** includes a first part **102** and a second part **104**. Each of the first part **102** and the second part **104** may define an internal volume configured to at least partially receive therein a hand, a wrist and/or one or more fingers of a user. In an exemplary embodiment, the first part **102** may be a fingerless glove having a body **103**. The body **103** includes a wrist portion **106** including a wrist opening **106***a*. The wrist opening **106***a* allows a hand "h" of a first user to be inserted into the first part **102**. The wrist portion **106** may partially cover the wrist of the user during use. The body **103** may enclose the hand "h" of the first user.

(34) The body **103** also includes five finger elements **108**, **110**, **112**, **114** and **116**. In some embodiments, the first finger element **108** may receive a thumb "t" (Latin pollex) of the first user. The first finger element **108** further defines a first end opening **108***a*, which allows the thumb "t" to extend therethrough. In an embodiment, the first finger element **108** may at least partly cover the thumb "t". In some embodiments, the second finger element **110** may receive an index finger "i" (Latin digitus secundus manus) of the first user. The second finger element **110** further defines a second end opening **110***a*, which allows the index finger "i" to extend therethrough. In an embodiment, the second finger element **110** may at least partly cover the index finger "i". In some embodiments, the third finger element **112** may receive a middle finger "m" (Latin digitus medius manus) of the first user. The third finger element **112** further defines a third end opening **112***a*, which allows the middle finger "m" to extend therethrough. In an embodiment, the third finger element **112** may at least partly cover the middle finger "m". In some embodiments, the fourth finger element **114** may receive a ring finger "r" (Latin digitus annularis manus) of the first user. The fourth finger element **114** further defines a fourth end opening **114***a*, which allows the ring finger "r" to extend therethrough. In an embodiment, the fourth finger element **114** may at least partly cover the ring finger "r". In some embodiments, the fifth finger element **116** may receive a little finger or pinky "l" (Latin digitus minimus manus) of the first user. The fifth finger element **116** further defines a fifth end opening **116***a*, which allows the little finger "l" to extend therethrough. In an embodiment, the fifth finger element **116** may at least partly cover the little finger "l".

(35) In some embodiments, the first part **102** may also define a dorsal or top section **118** and a palm or bottom section **120**. The dorsal section **118** may cover a dorsal part of the hand "h", while the palm section **120** may cover a palm of the hand "h". In an embodiment, the dorsal section **118** and the palm section **120** may be separately manufactured and then connected to each other by various methods, such as sewing, hot seal, adhesives, and so forth. In an alternative embodiment, the dorsal section **118** and the palm section **120** may be integrally manufactured.

(36) In an embodiment, the first part **102** may be made from a fabric-like material. In various embodiments, the material of the first part **102** may include a natural or synthetic fabric, wool, leather, rubber, latex, neoprene, and so forth. In another embodiment, the first part **102** may be made from a material having certain textures, durability and varying properties, such as waterproof or water resistant and easy to clean property. In yet another embodiment, the first part **102** may be made from a material which is flexible or elastic enough to allow the first part **102** to adapt to hands of different sizes. In an embodiment, the first part **102** may be designed in various sizes (i.e.,

small, medium, and large). In an exemplary embodiment, the first part **102** may be designed with an average size of an adult's hand. Alternatively, the first part **102** may be designed with an average size of a child's hand. In an exemplary embodiment, the first part **102** may be have a weight of 5-10 grams. Further, the first part **102** may comply with safety specifications of autism. In some embodiments, the material of the first part **102** may be able to withstand temperatures between −40° C. and +40° C. In an embodiment, the material of the first part **102** may include 100 percent soft cotton, with no labels. The material may be attuned to the sensory needs of children with autism spectrum disorder. In an embodiment, the material of the first part **102** may be selected such that it may be comfortable to wear for certain periods of time. Embodiments, as described above, are exemplary in nature and the first part **102** may be made from any suitable material as per requirements. In some embodiments, the first part **102** may have different colors incorporated onto the material to make the use of the apparatus **100**, fun for a child.
(37) In an exemplary embodiment, the second part **104** may also be a fingerless glove having a body **105**. The body **105** includes a wrist portion **122** including a wrist opening **122***a*. The wrist opening **122***a* receives a hand (not shown) of a second user. The wrist portion **122** may partially cover the wrist of the second user during use. The body **105** may enclose the hand of the second user.
(38) The body **105** also includes five finger elements **124**, **126**, **128**, **130** and **132**. In some embodiments, the first finger element **124** may receive a thumb of the second user. The first finger element **124** further defines a first end opening **124***a*, which allows the thumb to extend therethrough. In an embodiment, the first finger element **124** may at least partly cover the thumb. In some embodiments, the second finger element **126** may receive an index finger of the second user. The second finger element **126** further defines a second end opening **126***a*, which allows the index finger to extend therethrough. In an embodiment, the second finger element **126** may at least partly cover the index finger. In some embodiments, the third finger element **128** may receive a middle finger of the second user. The third finger element **128** further defines a third end opening **128***a*, which allows the middle finger to extend therethrough. In an embodiment, the third finger element **128** may at least partly cover the middle finger. In some embodiments, the fourth finger element **130** may receive a ring finger of the second user. The fourth finger element **130** further defines a fourth end opening **130***a*, which allows the ring finger to extend therethrough. In an embodiment, the fourth finger element **130** may at least partly cover the ring finger. In some embodiments, the fifth finger element **132** may receive a little finger of the second user. The fifth finger element **132** further defines a fifth end opening **132***a*, which allows the little finger to extend therethrough. In an embodiment, the fifth finger element **132** may at least partly cover the little finger.
(39) In an embodiment, the five finger elements **124**, **126**, **128**, **130** and **132** may be separately manufactured and connected to the body **105** by various methods, such as sewing, hot seal, adhesives, and so forth. In an alternative embodiment, the five finger elements **124**, **126**, **128**, **130** and **132** may be integrally manufactured with the body **105**.
(40) In some embodiments, the second part **104** may also define a dorsal section (not shown) and a palm section **134**. The dorsal section may cover a dorsal part of the hand, while the palm section **134** may cover a palm of the hand. In an embodiment, the dorsal section and the palm section **134** may be separately manufactured and then connected to each other by various methods, such as sewing, hot seal, adhesives, and so forth. In an alternative embodiment, the dorsal section and the palm section **134** may be integrally manufactured.
(41) In an embodiment, the second part **104** may be made from a fabric-like material. In various embodiments, the material of the second part **104** may include a natural or synthetic fabric, wool, leather, rubber, latex, neoprene, and so forth. In another embodiment, the second part **104** may be made from a material having certain textures, durability and varying properties, such as waterproof or water resistant and easy to clean property. In yet another embodiment, the second part **104** may

be made from a material which is flexible or elastic enough to allow the second part **104** to adapt to hands of different sizes. In an embodiment, the second part **104** may be designed in various sizes (i.e., small, medium, and large). In an exemplary embodiment, the second part **104** may be designed with an average size of a child's hand. Alternatively, the second part **104** may be designed with an average size of an adult's hand. In an exemplary embodiment, the second part **104** may be have a weight of 5-10 grams. Further, the second part **104** may comply with safety specifications of autism. In some embodiments, the material of the second part **104** may be able to withhold temperatures of between −40 degrees C. and +40 degrees C.

(42) In an embodiment, the material of the second part **104** may include 100 percent soft cotton, with no labels that is attuned to the sensory need of children with autism spectrum disorder. In an embodiment, the material of the second part **104** may be selected such that it may be comfortable to wear for certain periods of time. Embodiments, as described above, are exemplary in nature and second part **104** may be made from any suitable material as per requirements. In some embodiments, the second part **104** may have different colors incorporated onto the material to make the use of the apparatus **100** fun for a child. In an exemplary embodiment, the first part **102** may be attached to the second part **104** such that the palm section **120** of the first part **102** may abut the dorsal section of the second part **104**. In some embodiments, the sides of the first part **102** may be attached to the sides of the second part **104**. As shown in FIG. **1**C, an attachment region between the first part **102** and the second part **104** is highlighted by a reference numeral **136**. In an exemplary embodiment, the bodies **103** and **105** including the finger elements of each of the first part **102** and the second part **104** may be attached to each other, while the wrist portions **106** and **122** remain free. Therefore, the apparatus **100** provides minimum attachment between the first and second parts **102**, **104** to allow greater movement and flexibility. The first part **102** and the second part **104** may be attached to each other by any suitable attachment methods such as, but not limited to, sewing, heat seal, Velcro® and so forth. In an embodiment, the first user may be an adult or a guide, while the second user may be a child or a student. The child or the student may be suffering from a disease or a neurological disorder, such as autism. The apparatus **100** may allow the first user to control the movements of the second user. Further, the apparatus **100** may prevent or reduce skin to skin contact between the first user and the second user. Therefore, the apparatus **100** may be useful for training children/students who have high sensitivity to touch.

(43) In an exemplary embodiment, the apparatus **100** may provide an easy to use structure for the child and the adult. The apparatus **100** may assist the guide to teach the child essential self-care things. In an embodiment, the apparatus **100** may improve life and learning for a child. The apparatus **100** may be used in households, schools, training centers and so forth. In some embodiments, the apparatus **100** may be lightweight, with low packaging and shipping costs. Therefore, the apparatus **100** may be affordable and easily available to parents of learning disabled or autistic children. In an embodiment, the apparatus **100** may include a logo. In some embodiments, the apparatus **100** may be easily manufactured, reliable, cost effective, and durable. The apparatus **100** may be aesthetically pleasing to a child with autism spectrum disorder, or otherwise include design features having soothing or other beneficial effects for autistic persons. For example, an atypical color preference may be used, or a color coding adopted to promote learning, usage of visual patterns, etc.

(44) FIG. **2**A illustrates a bottom view of a hand-over-hand apparatus or device **200** (hereinafter referred to as “the apparatus **200**”) in use, in accordance with an embodiment of the present disclosure. FIG. **2**B illustrates a top view of the apparatus **200**. Referring to FIGS. **2**A and **2**B, the apparatus **200** may be a fingerless glove having a body **203**. The body **203** may define an internal volume configured to at least partially receive therein a hand, a wrist and/or one or more fingers of two users. The body **203** includes a wrist portion **206** including a wrist opening **206***a*. In an exemplary embodiment, the wrist opening **206***a* may have a suitable diameter which may allow two hands of the two users to be simultaneously received within the body **203**. In some

embodiments, the wrist portion **206** may partially cover the wrist of both the users during use. The body **203** may also have suitable dimensions to enclose the hands of both the users.
(45) The body **203** also includes five finger elements **208**, **210**, **212**, **214** and **216**. In some embodiments, the first finger element **208** may receive a thumb of the each of the users. The first finger element **208** further defines a first end opening **208***a*, which may allow the thumb of each user to extend therethrough. In an embodiment, the first finger element **208** may at least partly cover the thumbs of both the users. In some embodiments, the second finger element **210** may receive an index finger of each of the users. The second finger element **210** further defines a second end opening **210***a*, which allows the index fingers of both the users to extend therethrough. In an embodiment, the second finger element **210** may at least partly cover the index fingers of both the users. In some embodiments, the third finger element **212** may receive a middle finger of each of the users. The third finger element **212** further defines a third end opening **212***a*, which allows the middle finger of each of the users to extend therethrough. In an embodiment, the third finger element **212** may at least partly cover the middle fingers of both the users. In some embodiments, the fourth finger element **214** may receive a ring finger of each of the users. The fourth finger element **214** further defines a fourth end opening **214***a*, which allows the ring finger of each of the users to extend therethrough. In an embodiment, the fourth finger element **214** may at least partly cover the ring fingers of both the users. In some embodiments, the fifth finger element **216** may receive a little finger of each of the users. The fifth finger element **216** further defines a fifth end opening **216***a*, which allows the little finger of each user to extend therethrough. In an embodiment, the fifth finger element **216** may at least partly cover the little fingers of both the users.
(46) In an embodiment, the five finger elements **208**, **210**, **212**, **214** and **216** may be separately manufactured and connected to the body **203** by various methods, such as sewing, hot seal, adhesives, and so forth. In an alternative embodiment, the five finger elements **208**, **210**, **212**, **214** and **216** may be integrally manufactured with the body **203**.
(47) In some embodiments, the apparatus **200** may also define a dorsal section **218** and a palm section **220**. The dorsal section **218** may cover a dorsal part of the hand, while the palm section **220** may cover a palm of the hand. In an embodiment, the dorsal section **218** and the palm section **220** may be separately manufactured and then connected to each other by various methods, such as sewing, hot seal, adhesives, and so forth. In an alternative embodiment, the dorsal section **218** and the palm section **220** may be integrally manufactured.
(48) In an embodiment, the apparatus **200** may be made from a fabric-like material. In various embodiments, the material of the apparatus **200** may include a natural or synthetic fabric, wool, leather, rubber, latex, neoprene, and so forth. In another embodiment, the apparatus **200** may be made from a material having certain textures, durability and varying properties, such as waterproof or water resistant and easy to clean property. In yet another embodiment, the apparatus **200** may be made from a material which is flexible or elastic enough to allow the apparatus **200** to receive the hands of two users simultaneously. Embodiments, as described above, are exemplary in nature and the apparatus **200** may be made from any suitable material as per requirements. In an embodiment, a first user may be an adult or a guide, while a second user may be a child or a student suffering from a disease or a neurological disorder, such as autism. The apparatus **200** may allow the first user to control the movements of the second user. Therefore, the apparatus **200** may allow the first user to teach the second user various skills such as, but not limited to, self-care tasks, handwriting skills, scissors skills, hand motor coordination and so forth.
(49) FIG. **3**A illustrates a bottom perspective view of a hand-over-hand apparatus or device **300** (hereinafter referred to as “the apparatus **300**”), in accordance with an embodiment of the present disclosure. FIG. **3**B illustrates a side perspective view of the apparatus **300** in use. FIG. **3**C illustrates a bottom perspective view of the apparatus **300** in use. FIG. **3**D illustrates a top view of the apparatus **300**. Referring to FIGS. **3**A-**3**D, the apparatus **300** may include a glove **302** having a body **303**. The body **303** includes a wrist portion **306** including a wrist opening **306***a*. The wrist

opening **306***a* allows a hand “h**1**” of a first user to be inserted into the glove **302**. The wrist portion **306** may partially cover the wrist of the first user during use. The body **303** may enclose the hand “h**1**” of the first user.

(50) The body **303** also includes five finger elements **308**, **310**, **312**, **314** and **316**. In some embodiments, the first finger element **308** may receive a thumb of the first user. In an exemplary embodiment, the first finger element **308** may define an enclosed volume configured to enclose the thumb of the first user. In some embodiments, the second finger element **310** may receive an index finger of the first user. In an exemplary embodiment, the second finger element **310** may define an enclosed volume configured to enclose the index finger of the first user. In some embodiments, the third finger element **312** may receive a middle finger of the first user. In an exemplary embodiment, the third finger element **312** may define an enclosed volume configured to enclose the middle finger of the first user. In some embodiments, the fourth finger element **314** may receive a ring finger of the first user. In an exemplary embodiment, the fourth finger element **314** may define an enclosed volume configured to enclose the ring finger of the first user. In some embodiments, the fifth finger element **316** may receive a little finger of the first user. In an exemplary embodiment, the fifth finger element **316** may define an enclosed volume configured to enclose the little finger of the first user.

(51) In an embodiment, the five finger elements **308**, **310**, **312**, **314** and **316** may be separately manufactured and connected to the body **303** by various methods, such as sewing, hot seal, adhesives, and so forth. In an alternative embodiment, the five finger elements **308**, **310**, **312**, **314** and **316** may be integrally manufactured with the body **303**.

(52) In some embodiments, the glove **302** may also define a dorsal section **318** and a palm section **320**. The dorsal section **318** may cover a dorsal part of the hand “h**1**”, while the palm section **320** may cover a palm of the hand “h**1**”. In an embodiment, the dorsal section **318** and the palm section **320** may be separately manufactured and then connected to each other by various methods, such as sewing, hot seal, adhesives, and so forth. In an alternative embodiment, the dorsal section **318** and the palm section **320** may be integrally manufactured.

(53) In an embodiment, the glove **302** may be made from a fabric-like material. In various embodiments, the material of the glove **302** may include a natural or synthetic fabric, wool, leather, rubber, latex, neoprene, and so forth. In another embodiment, the glove **302** may be made from a material having certain textures, durability and varying properties, such as waterproof or water resistant and easy to clean property. In yet another embodiment, the glove **302** may be made from a material which is flexible or elastic enough to allow the glove **302** to accommodate hands of different sizes. Embodiments, as described above, are exemplary in nature and the glove **302** may be made from any suitable material as per requirements.

(54) In an exemplary embodiment, the apparatus **300** may also include five material bands **322**, **324**, **326**, **328**, and **330** attached to the corresponding five finger elements **308**, **310**, **312**, **314** and **316** of the glove **302**. The material of the material bands may include an elastic material and/or a fabric material. The material bands **322**, **324**, **326**, **328**, and **330** may be attached to the glove **302** in a direction “D” relative to the palm section **320**. In an exemplary embodiment, the material band **322** may be attached to the first finger element **308** by any suitable attachment methods, such as, but not limited to, sewing, hot seal, Velcro®, adhesive and so forth. In some embodiments, the material band **322** may be attached at sides of the first finger element **308**. Further, the first material band **322** may form a closed loop in the direction “D”. Therefore, the first material band **322** may extend beneath the first finger element **308**. In another embodiment, the first material band **322** may form a fully closed loop around the first finger element **308**. The first material band **322** may also be reversed around the first finger element **308**, so that the first material band **322** extends in a direction opposite to the direction “D”. Therefore, in the reverse configuration, the first material band **322** may extend on top of the first finger element **308**. In an exemplary embodiment, the first material band **322** may receive a thumb of a second user. In a further embodiment, the first material

band **322** may allow the second user to slidably insert the thumb beneath the thumb of the first user. In an alternative embodiment, the first material band **322** may allow the second user to slidably insert the thumb on top of the thumb of the first user.

(55) In an exemplary embodiment, the second material band **324** may be attached to the second finger element **310** by any suitable attachment methods, such as, but not limited to, sewing, hot seal, Velcro® hook and loop fasteners, adhesives and so forth. In some embodiments, the second material band **324** may be attached at sides of the second finger element **310**. Further, the second material band **324** may form a closed loop in the direction “D”. Therefore, the second material band **324** may extend beneath the first finger element **308**. In an exemplary embodiment, the second material band **324** may receive an index finger of the second user. In a further embodiment, the second material band **324** may allow the second user to slidably insert the index finger beneath the index finger of the first user. In some embodiments, the second material band **324** may form a fully closed loop around the second finger element **310**. The second material band **324** may also be reversed around the second finger element **310**, so that the second material band **324** extends in a direction opposite to the direction “D”. Therefore, in the reverse configuration, the second material band **324** may extend on top of the second finger element **310**. In an embodiment, the second material band **324** may allow the second user to position the index finger on top of the index finger of the first user.

(56) In an exemplary embodiment, the third material band **326** may be attached to the third finger element **312** by any suitable attachment methods, such as, but not limited to, sewing, hot seal, Velcro®, adhesive and so forth. In some embodiments, the third material band **326** may be attached at sides of the third finger element **312**. Therefore, the third material band **326** may form a closed loop in the direction “D”. Further, the third material band **326** may extend beneath the third finger element **312**. In an exemplary embodiment, the third material band **326** may receive a middle finger of the second user. In a further embodiment, the third material band **326** may allow the second user to slidably insert the middle finger beneath the middle finger of the first user. In some embodiments, the third material band **326** may form a fully closed loop around the third finger element **312**. The third material band **326** may also be reversed around the third finger element **312**, so that the third material band **326** extends in a direction opposite to the direction “D”. Therefore, in the reverse configuration, the third material band **326** may extend on top of the third finger element **312**. As a result, the third material band **326** may also allow the second user to position the middle finger on top of the middle finger of the first user.

(57) In an exemplary embodiment, the fourth material band **328** may be attached to the fourth finger element **314** by any suitable attachment methods, such as, but not limited to, sewing, hot seal, Velcro®, adhesive and so forth. In some embodiments, the fourth material band **328** may be attached at sides of the fourth finger element **314**. Therefore, the fourth material band **328** may form a closed loop in the direction “D”. Further, the fourth material band **328** may extend beneath the fourth finger element **314**. In an exemplary embodiment, the fourth material band **328** may receive a ring finger of the second user. In a further embodiment, the fourth material band **328** may allow the second user to slidably insert the ring finger beneath the ring finger of the first user. In some embodiments, the fourth material band **328** may form a fully closed loop around the fourth finger element **314**. Therefore, the fourth material band **328** can be reversed around the fourth finger element **314** in order to allow the second user to position the ring finger on top of the ring finger of the first user.

(58) In an exemplary embodiment, the fifth material band **330** may be attached to the fifth finger element **316** by any suitable attachment methods, such as, but not limited to, sewing, hot seal, Velcro®, adhesive and so forth. In some embodiments, the fifth material band **330** may be attached at sides of the fifth finger element **316**. Therefore, the fifth material band **330** may form a closed loop in the direction “D”. Further, the fifth material band **330** may extend beneath the fifth finger element **316**. In an exemplary embodiment, the fifth material band **330** may receive a little finger of

the second user. In a further embodiment, the fifth material band **330** may allow the second user to slidably insert the little finger beneath the little finger of the first user. In some embodiments, the fifth material band **330** may form a fully closed loop around the fifth finger element **316**. Therefore, the fifth material band **330** may be reversed around the fifth finger element **316** to allow the second user to position the little finger on top of the little finger of the first user.
(59) In various embodiments, a material of each of the material bands **322**, **324**, **326**, **328**, and **330** includes an elastic fabric, rubber, latex, neoprene, and so forth.
(60) As illustrated in FIGS. **3**A-**3**D, the material bands **322**, **324**, **326**, **328**, and **330** may be configured to secure the hand "h**2**" of the second user under the hand "h**1**" of the first user. Alternatively, the material bands **322**, **324**, **326**, **328**, and **330** may also be configured to secure the hand "h**2**" of the second user on top of the hand "h**1**" of the first user.
(61) In an embodiment, the first user may be an adult or a guide, while the second user may be a child or a student. The apparatus **300** may allow the first user to control the movements of the second user. Therefore, the apparatus **300** allows the first user to teach the second user various skills such as, but not limited to, self-care tasks, handwriting skills, scissors skills, hand motor coordination and so forth.
(62) In alternative embodiments, the first user may wear the material bands **322**, **324**, **326**, **328**, and **330**, while the second user may wear the glove **302**.
(63) FIG. **4**A illustrates a perspective view of a hand-over-hand apparatus or device **400** (hereinafter referred to as "the apparatus **400**"), in accordance with an embodiment of the present disclosure. FIG. **4**B illustrates a side perspective view of the apparatus **400** in use. FIG. **4**C illustrates another side perspective view of the apparatus **400** in use. FIG. **4**D illustrates a top view of the apparatus **400** in use. The apparatus **400** may be similar to the apparatus **100** (as shown in FIGS. **1**A-**1**C). Referring to FIGS. **4**A-**4**D, the apparatus **400** includes a first part **402** and a second part **404**. In an exemplary embodiment, the first part **402** may be a fingerless glove having a body **403**. The body **403** includes a wrist portion **406** including a wrist opening **406***a*. The wrist opening **406***a* receives a hand "h**6**" of a first user. The wrist portion **406** may partially cover the wrist of the user in use. The body **403** may enclose the hand "h**6**" of the first user. The body **403** also includes five finger elements **408**, **410**, **412**, **414** and **416**. In some embodiments, the first finger element **408** may receive a thumb "t**1**" of the first user. The first finger element **408** further defines a first end opening **408***a*, which allows the thumb "t**1**" to extend therethrough. In an embodiment, the first finger element **408** may at least partly cover the thumb "t**1**". In some embodiments, the second finger element **410** may receive an index finger "i**1**" of the first user. The second finger element **410** further defines a second end opening **410***a*, which allows the index finger "i**1**" to extend therethrough. In an embodiment, the second finger element **410** may at least partly cover the index finger "i**1**". In some embodiments, the third finger element **412** may receive a middle finger "m**1**" of the first user. The third finger element **412** further defines a third end opening **412***a*, which allows the middle finger "m**1**" to extend therethrough. In an embodiment, the third finger element **412** may at least partly cover the middle finger "m**1**". In some embodiments, the fourth finger element **414** may receive a ring finger "r**1**" of the first user. The fourth finger element **414** further defines a fourth end opening **414***a*, which allows the ring finger "r**1**" to extend therethrough. In an embodiment, the fourth finger element **414** may at least partly cover the ring finger "r**1**". In some embodiments, the fifth finger element **416** may receive a little finger "l**1**" of the first user. The fifth finger element **416** further defines a fifth end opening **416***a*, which allows the little finger "l**1**" to extend therethrough. In an embodiment, the fifth finger element **416** may at least partly cover the little finger "l**1**". In some embodiments, the first part **402** may also define a dorsal section **418** and palm section (not shown).
(64) In an embodiment, the first part **402** may be made from a fabric-like material. In another embodiment, the first part **402** may be made from a material having certain textures, durability and varying properties, such as waterproof or water resistant and easy to clean property. In yet another

embodiment, the first part **402** may be made from a material which is flexible enough to allow the first part **402** to adapt to hands of different sizes. Embodiments, as described above, are exemplary in nature and the first part **402** may be made from any suitable material as per requirements.
(65) In an exemplary embodiment, the second part **404** may also be a fingerless glove having a body **405**. The body **405** includes a wrist portion **422** including a wrist opening **422***a*. The wrist opening **422***a* receives a hand "h**7**" of a second user. The wrist portion **422** may partially cover the wrist of the second user during use. The body **405** may enclose the hand "h**7**" of the second user. The body **405** also includes five finger elements **424**, **426**, **428**, **430** and **432**. In some embodiments, the first finger element **424** may receive a thumb "t**2**" of the second user. The first finger element **424** further defines a first end opening **424***a*, which allows the thumb "t**2**" to extend therethrough. In an embodiment, the first finger element **424** may at least partly cover the thumb "t**2**". In some embodiments, the second finger element **426** may receive an index finger "i**2**" of the second user. The second finger element **426** further defines a second end opening **426***a*, which allows the index finger "i**2**" to extend therethrough. In an embodiment, the second finger element **426** may at least partly cover the index finger "i**2**". In some embodiments, the third finger element **428** may receive a middle finger "m**2**" of the second user. The third finger element **428** further defines a third end opening **428***a*, which allows the middle finger "m**2**" to extend therethrough. In an embodiment, the third finger element **428** may at least partly cover the middle finger "m**2**". In some embodiments, the fourth finger element **430** may receive a ring finger "r**2**" of the second user. The fourth finger element **430** further defines a fourth end opening **430***a*, which allows the ring finger "r**2**" to extend therethrough. In an embodiment, the fourth finger element **430** may at least partly cover the ring finger "r**2**". In some embodiments, the fifth finger element **432** may receive a little finger "l**2**" of the second user. The fifth finger element **432** further defines a fifth end opening **432***a*, which allows the little finger "l**2**" to extend therethrough. In an embodiment, the fifth finger element **432** may at least partly cover the little finger "l**2**". In some embodiments, the second part **404** may also define a dorsal section (not shown) and palm section **434**.
(66) In an embodiment, the second part **404** may be made from a fabric-like material. In another embodiment, the second part **404** may be made from a material having certain textures, durability and varying properties, such as waterproof or water resistant and easy to clean property. In yet another embodiment, the second part **404** may be made from a material which is flexible enough to accommodate hands of different sizes. Embodiments, as described above, are exemplary in nature and the second part **404** may be made from any suitable material as per requirements.
(67) In an exemplary embodiment, the first part **402** may be attached to the second part **404** such that the palm section of the first part **402** may abut the dorsal section of the second part **404**. In some embodiments, the sides of the first part **402** may be attached to the sides of the second part **404**. As illustrated in FIG. **4**D, an attachment region between the first part **402** and the second part **104** is shown by a reference numeral **436**. As can be clearly seen in FIG. **4**D, the finger elements **408**, **410**, **412**, **414** and **416** of the first part **402** are attached to the corresponding finger elements **424**, **426**, **428**, **430** and **432** of the second part **404**. Further, the wrist portion **406** of the first part **402** may also be attached to the wrist portion **422** of the second part **404**. Therefore, the apparatus **400** provides increased attachment between the first and second parts **402**, **404**, thereby allowing greater control. The first part **402** and the second part **404** may be attached to each other by any suitable attachment methods such as, but not limited to, sewing, heat seal, Velcro® and so forth. In an embodiment, the first user may be an adult or a guide, while the second user may be a child or a student suffering from a disease or a neurological disorder, such as autism. The apparatus **400** may allow the first user to control the movements of the second user. Further, the apparatus **400** may prevent or reduce skin to skin contact between the first user and the second user.
(68) In an embodiment, lengths of the finger elements **408**, **410**, **412**, **414** and **416** of the first part **402** may be greater than lengths of the corresponding finger elements **108**, **110**, **112**, **114** and **116** of the first part **102** of the apparatus **100** (shown in FIGS. **1**A to **1**C). Similarly, lengths of the finger

elements **424**, **426**, **428**, **430** and **432** of the second part **404** may be greater than lengths of the corresponding finger elements **124**, **126**, **128**, **130** and **132** of the second part **104** of the apparatus **100**. Further, the first part **402** and the second part **404** may be attached to each other at the sides of the corresponding finger elements and at the corresponding wrist portions **406** and **422**. On the other hand, the first and second parts **102** and **104** of the apparatus **100** are connected only at the sides, and the wrist portions **106** and **122** are free from each other. Therefore, the apparatus **400** includes greater attachment between the first and second parts **402**, **404**, thereby providing greater control. On the other hand, the apparatus **100** provides lesser attachment to allow greater relative movement and flexibility between the first and second parts **102**, **104**.

(69) FIG. **5**A illustrates a bottom perspective view of a hand-over-hand apparatus or device **500** (hereinafter referred to as “the apparatus **500**”), in accordance with an embodiment of the present disclosure. FIG. **5**B illustrates a bottom perspective view of the apparatus **500** with one hand inserted therein. FIG. **5**C illustrates a top view of the apparatus **500**. In an exemplary embodiment, the apparatus **500** may include a first glove **502** and a second glove **504**. The first glove **502** may be configured to receive a hand of a first user. The first glove **502** may include a body **503**, a wrist portion **506**, and five finger elements **508**, **510**, **512**, **514** and **516**. The wrist portion **506** includes a wrist opening **506***a*. The wrist opening **506***a* allows the hand of the first user to be inserted within the body **503**. The wrist portion **506** may partially cover the wrist of the first user during use. The body **503** may enclose the hand of the first user. In some embodiments, the body **503** may include the five finger elements **508**, **510**, **512**, **514** and **516**. In some embodiments, the first finger element **508** may receive a thumb of the first user. In an exemplary embodiment, the first finger element **508** may define an enclosed volume configured to enclose the thumb of the first user. In some embodiments, the second finger element **510** may receive an index finger of the first user. In an exemplary embodiment, the second finger element **510** may define an enclosed volume configured to enclose the index finger of the first user. In some embodiments, the third finger element **512** may receive a middle finger of the first user. In an exemplary embodiment, the third finger element **512** may define an enclosed volume configured to enclose the middle finger of the first user. In some embodiments, the fourth finger element **514** may receive a ring finger of the first user. In an exemplary embodiment, the fourth finger element **514** may define an enclosed volume configured to enclose the ring finger of the first user. In some embodiments, the fifth finger element **516** may receive a little finger of the first user. In an exemplary embodiment, the fifth finger element **516** may define an enclosed volume configured to enclose the little finger of the first user. In some embodiments, the first glove **502** may also define a dorsal section **518** and a palm section **520**.

(70) In an embodiment, the first glove **502** may be made from a fabric-like material. In another embodiment, the first glove **502** may be made from a material having certain textures, durability and varying properties, such as waterproof or water resistant and easy to clean property. In yet another embodiment, the first glove **502** may be made from a material which is flexible enough to allow the first glove **502** to accommodate hands of different sizes.

(71) In an exemplary embodiment, the second glove **504** may be similar to the first glove **502**. Therefore, similar parts are not described in detail hereinafter. The second glove **504** may be configured to receive a hand of a second user. The second glove **504** may include a body **505**, a wrist portion **522**, and five finger elements **524**, **526**, **528**, **530** and **532**. The wrist portion **522** includes a wrist opening **522***a*. The wrist opening **522***a* allows the hand of the second user to be inserted within the second glove **504**. The wrist portion **522** may partially cover the wrist of the second user during use. In an exemplary embodiment, the five finger elements **524**, **526**, **528**, **530** and **532** of the second glove **504** are similar to the corresponding five finger elements **508**, **510**, **512**, **514** and **516** of the first glove **502**. The five finger elements **524**, **526**, **528**, **530** and **532** of the second glove **504** may enclose a thumb, an index finger, a middle finger, a ring finger and a little finger, respectively, of the second user. In some embodiments, the second glove **504** may also define a dorsal section (not shown) and a palm section **534**.

(72) In an embodiment, the second glove **504** may be made from a fabric-like material. In another embodiment, the second glove **504** may be made from a material having certain textures, durability and varying properties, such as waterproof or water resistant and easy to clean property. In yet another embodiment, the second glove **504** may be made from a material which is flexible enough to allow the second glove **504** to adapt to hands of different sizes.
(73) In an exemplary embodiment, the first glove **502** may be attached to the second glove **504** such that the palm section **520** of the first glove **502** may abut the dorsal section of the second glove **504**. In some embodiments, sides of the first glove **502** may be attached to sides of the second glove **504**. Further, each finger element **508**, **510**, **512**, **514** and **516** of the first glove **502** may be attached to the corresponding finger elements **524**, **526**, **528**, **530** and **532** of the second glove **504**. As shown in FIG. 5C, an attachment region between the first glove **502** and the second glove **504** is shown by a reference numeral **536**. In some embodiments, the wrist portion **506** of the first glove **502** may also be attached to the wrist portion **522** of the second glove **504**. Therefore, the apparatus **500** provides increased attachment to allow greater control. Further, the apparatus **500** may completely prevent skin to skin interaction between the first user and the second user. The first glove **502** and the second glove **504** may be attached to each other by any suitable attachment methods such as, but not limited to, sewing, heat seal, Velcro® and so forth. In an embodiment, the first user may be an adult or a guide, while the second user may be a child or a student. The apparatus **500** may allow the first user to control the movements of the second user. Alternatively, the first user may be a child or a student, while the second user may be an adult or a guide.
(74) FIG. **6**A illustrates a side perspective view of a hand-over-hand apparatus or device **600** (hereinafter referred to as “the apparatus **600**”) in use, in accordance with an embodiment of the present disclosure. FIG. **6**B illustrates a bottom perspective view of the apparatus **600**. FIG. **6**C illustrates a top view of the apparatus **600**. The apparatus **600** may be similar to the apparatus **500** (as shown in FIGS. **5**A-**5**C). Further, for the purposes of the present disclosure, similar parts are provided with corresponding reference numerals.
(75) In an exemplary embodiment, the apparatus **600** may include a first glove **602** and a second glove **604**. The first glove **602** may be configured to enclose a hand “h**3**” of a first user. The first glove **602** may include a body **603**, a wrist portion **606**, and five finger elements **608**, **610**, **612**, **614** and **616**. The wrist portion **606** includes a wrist opening (not shown). In some embodiments, the wrist opening allows the hand “h**3**” of the first user to be inserted within the body **603**. The wrist portion **606** may partially cover the wrist of the first user during use. The body **603** may enclose the hand “h**3**” of the first user. In some embodiments, the body **603** may include the five finger elements **608**, **610**, **612**, **614** and **616**. The five finger elements **608**, **610**, **612**, **614** and **616** of the first glove **602** may enclose a thumb, an index finger, a middle finger, a ring finger and a little finger, respectively, of the first user. In some embodiments, the first glove **602** may also define a dorsal section **618** and a palm section **620**.
(76) In an embodiment, the first glove **602** may be made from a fabric-like material. In another embodiment, the first glove **602** may be made from a material having certain textures, durability and varying properties, such as waterproof or water resistant and easy cleanability. In yet another embodiment, the first glove **602** may be made from a material which is flexible enough to allow the first glove **602** to accommodate hands of different sizes.
(77) In an exemplary embodiment, the second glove **604** may be similar to the first glove **602**. The second glove **604** may be configured to receive a hand “h**4**” of the second user. The second glove **604** may include a body **605**, a wrist portion **622**, and five finger elements **624**, **626**, **628**, **630** and **632**. The wrist portion **622** includes a wrist opening (not shown). The wrist opening receives the hand “h**4**” of the second user. The wrist portion **622** may partially cover the wrist of the first user during use. In an exemplary embodiment, the five finger elements **624**, **626**, **628**, **630** and **632** of the second glove **604** are similar to the corresponding five finger elements **608**, **610**, **612**, **614** and **616** of the first glove **602**. The five finger elements **624**, **626**, **628**, **630** and **632** of the second glove

604 may enclose a thumb, an index finger, a middle finger, a ring finger and a little finger, respectively, of the second user. In some embodiments, the second glove **604** may also define a dorsal section (not shown) and a palm section **634**. In an embodiment, the second glove **604** may be made from a fabric-like material. In another embodiment, the second glove **604** may be made from a material having certain textures, durability and varying properties such as, waterproof or water resistant and easy to clean property. In yet another embodiment, the second glove **604** may be made from a material which is flexible enough to allow the second glove **604** to adapt to hands of different sizes.

(78) In an exemplary embodiment, the first glove **602** may be attached to the second glove **604** such that the palm section **620** of the first glove **602** may face the dorsal section of the second glove **604**. In an exemplary embodiment, the first glove **602** and the second glove **604** may only be attached at tips of the corresponding finger elements of the first glove **602** and the second glove **604**. As illustrated in FIG. **6**C, an attachment region of the first glove **602** and the second glove **604** is highlighted by a reference numeral **636**. In an exemplary embodiment, the apparatus **600** may provide maximum movement and flexibility to each of the first user and the second user. Further, the apparatus **600** may completely prevent skin to skin interaction between the first user and the second user. Therefore, the apparatus **600** may be useful for training children/students who have high sensitivity to touch. Further, children/students having no experience of touching an external object (e.g., a pair of scissors) may also be trained with the apparatus **600**. The first glove **602** and the second glove **604** may be attached to each other by any suitable attachment methods such as, but not limited to, sewing, heat seal, Velcro® and so forth. In some embodiments, additional attachments can also be provided by any suitable methods such as, but not limited to, Velcro®, to allow a desired control over hand movements. In an embodiment, the first user may be an adult or a guide, while the second user may be a child or a student. The apparatus **600** may allow the first user to control the movements of the second user. Alternatively, the first user may be a child or a student, while the second user may be an adult or a guide.

(79) The first and second gloves **602**, **604** may be attached only at the tips of the corresponding finger elements to allow maximum movement and flexibility to each of the first user and the second user. On the other hand, the first and second gloves **502**, **504** (shown in FIGS. **5**A to **5**C) may be attached at the corresponding finger elements as well as the corresponding wrist portions to allow greater control.

(80) In an embodiment, the first and second gloves **602**, **604** may be made of a breathable fabric that minimizes sweating due to prolonged use.

(81) FIGS. **7**A and **7**B illustrate various views of an apparatus **700**, in accordance with an aspect of the present disclosure. The apparatus **700** may be similar to the apparatus **600**. The apparatus **700** includes a first glove **702** and a second glove **704**. The first glove **702** includes a body **703**, a wrist portion **706**, and five finger elements **708**. The second glove **704** includes a body **710**, and five finger elements **714**. The use of the apparatus **700** may be similar to the apparatus **600**. As illustrated in FIGS. **7**A and **7**B, during use, the first glove **702** can be moved to control movements of the second glove **704** due to attachment between the first and second gloves **702**, **704**. Specifically, each of the finger elements **708** of the first glove **702** is attached to a corresponding element **714** of the second glove **704**. Therefore, the apparatus **700** allows a first user wearing the first glove **702** to control individual movements of the fingers of a second user wearing the second glove **704**. Further, the body **703** of the first glove **702** may be also attached to the body **710** of the second glove **704**, thereby allowing the first user to control the movements and apply pressure on the hand of the second user.

(82) FIG. **8** illustrates a top perspective view of a hand-over-hand device **800**, in accordance with an embodiment of the present disclosure. The apparatus **800** may be similar to the apparatus **300**, except for an additional material band that holds palms of the users together. The material of the material band may be an elastic material and/or a fabric material. Further, for the purposes of the

present disclosure, similar parts are provided with corresponding reference numerals.
(83) The apparatus **800** may include a glove **802** having a body **803**. The body **803** may enclose a hand of the first user. The body **803** also includes five finger elements **808**, **810**, **812**, **814** and **816**. The finger elements **808**, **810**, **812**, **814** and **816** may define enclosed volumes configured to enclose a thumb, an index finger, a middle finger, a ring finger and a little finger, respectively, of a first user.
(84) In an exemplary embodiment, the apparatus **800** may also include six material bands **822**, **824**, **826**, **828**, **830**, and **832** attached to the corresponding five finger elements **808**, **810**, **812**, **814 816** and the body **803** of the glove **802**. In an exemplary embodiment, the five material bands **822**, **824**, **826**, **828**, **830** may receive a thumb, an index finger, a middle finger, a ring finger and a little finger, respectively, of a second user, while the material band **832** may support the hand the second user. Specifically, the material band **832** may secure the palm of the second user to the palm of the first user. The material bands **822**, **824**, **826**, **828**, **830**, and **832** may be configured to secure the hand of the second user under the hand of the first user. Alternatively, the material bands **822**, **824**, **826**, **828**, **830**, and **832** may also be configured to secure the hand of the second user on top of the hand of the first user.
(85) The apparatus **800** may include six material bands **822**, **824**, **826**, **828**, **830**, and **832** configured to secure all the fingers and the hand of the second user under the hand of the first user to allow greater control and provide maximum pressure. On the other hand, the apparatus **300** includes five material bands **322**, **324**, **326**, **328**, and **330** configured to secure only the fingers of the second user to the fingers of the first user. Therefore, the apparatus **300** provides maximum movement and flexibility to each of the first user and the second user.
(86) The apparatus **100**, as described above, can be used to perform various operations, such as, but not limited to, learning and training of disabled or autistic children. Such operations can also be performed by any of the apparatus **200**, **300**, **400**, **500**, **600**, **700**, or **800**.
(87) Exemplary methods of using the apparatus **100**, as an educational and learning tool through tactile sensory demonstration and action, are described below. Such exemplary methods may be communicated to guides/teachers/parents in the form of detailed instructions to facilitate training of students/children with developmental disabilities.
(88) For these operations, the guide or the adult will need to stand behind the student or child (collectively “student”), while using the apparatus with the student. Tasks are to be practiced with the student consistently over time. In an example, the first part **102** of the apparatus **100** is worn by the guide, while the second part **104** is worn by the student.
(89) All tasks are described in beginning, middle and end phases. These phases may be just a template for the guide to use. The phases do not have any particular length of time and are not necessarily mutually exclusive of one another. The guide must be mindful in assessing the needs of the student, i.e., whether the guide needs to: slow down in his/her approach, apply more or less physical strength, when and when not to use verbal prompts, when to fade out verbal prompts, and when to test the student.
(90) The guide must understand and keep in mind that the apparatus **100** is only a learning tool, and that the guide himself must aid in the learning experience. The guide must learn to challenge the student without overwhelming the student. The guide should use positive reinforcement throughout the learning experience. Some students respond well to verbal praise. Others may need more, such as a reward after a specific number of attempts (e.g., a treat, sing a song, watch a video, etc.).
(91) Results of the teaching and education method are contingent upon a frequency and consistency of use/practice of the apparatus **100**. As in any other type of learning or therapy, the more it is applied, the greater the possibilities of learning, and at a faster rate.
(92) The length of a session should be specific to the student's capabilities. If a session proves to be too long or overwhelming for the student, then the student may become disinterested and/or disengaged. The guide must be mindful of the student's mental status at all times. It may be optimal

to begin with just a few minutes and add to session time, as the learning progresses. If time allows, the guide may attempt multiple times during a given day, at different times of the day. A recommended use is every day, during the beginning of a task. If this is not possible, then the recommended minimum usage is every other day.

(93) As described above, all tasks are divided into the beginning, middle and end phases. During the beginning phase, the task is introduced.

(94) During the beginning phase of these operations, it is paramount that the guide:

(95) 1. Use his/her own strength to perform task.

(96) 2. Exaggerate the motion, i.e., moving as slowly as possible, so that the student can feel the strength being applied and understand the focus needed.

(97) 3. Use verbal declaratives, such as: “wow, this is really hard to open”, “let's do this”. Such language may take the pressure off of the student to perform (no judgment) and enable the student to share the experience with the guide. It is important to not judge the student during the learning experience, so that the student's learning is not contingent upon student wanting to please the guide. Verbal prompts, should be used as needed and only one at a time, so as not to overwhelm the student. It is important not to use verbal prompts in excess, as they can also distract or confuse the student.

(98) The middle phase involves an increase in the student's motivation to act.

(99) During the middle phase:

(100) 1. The guide begins to decrease his/her own strength in the task and allows the student to begin to apply his/her own strength.

(101) 2. As the student gains strength in the task, the guide reduces use of verbal prompts.

(102) After the first few trials (beginning phase), the guide may begin to lessen his/her strength (middle phase) used in the task, allowing the student to become engaged in the task. For example, if required, while hands are on a jar, the guide can just wait until the student makes an effort to twist the lid. The guide should allow the student to try as much as he/she can, before the guide joins in to help. The guide applies as much strength as needed, but less and less as time goes on, thereby allowing the student to take on more responsibility for the task.

(103) If the student is slow to move, the guide needs to be patient and allow the student all the time he/she needs. The guide may add some declarative prompts, such as: “let's open this”, “help me open the jar”. However, if the guide chooses to use verbal prompts, it is important to state it only once and wait for the student to respond.

(104) If the student is not motivated to engage in the task, then a reinforcement may be added. The reinforcement may be something that the student or child enjoys. For example, the reinforcement can be a small treat to eat, or a quick online video. The guide can state: “after we open this jar, you can watch a video” or “eat a treat”.

(105) After several days of trials (depending on the student's ability to master the task), the guide can begin to increase the variety and number of jars/bottles/tubes (whatever is being worked on). This may help the student to generalize the task quicker (versus learning only on one jar at a time).

(106) As the student gains the strength needed and the ability to open jars, bottles, containers, etc., the guide relinquishes his/her own strength, allowing the student to do all the work. Once the guide believes that the student is relying on his/her own strength, it is then an opportune time for the guide to present the jars/bottles, etc. to the student without the use of the apparatus **100**, i.e., testing mastery of the task.

(107) The end phase may involve testing the student's mastery of the task.

(108) The end phase includes:

(109) 1. At this point, the guide believes that the student is using his/her own strength to complete task, and guide is using none.

(110) 2. The guide can begin to test the student without the apparatus **100**.

(111) If the student is able to perform the task several times, without the aid of the apparatus **100**,

then he/she has mastered the task.

(112) The guide is to provide verbal praise, and perhaps a reward, if the guide believes it is deserved. Some students may require more praise than others; this would depend on the guide's assessment of the student. At this point, it may be no longer necessary to practice this task continually. However, the guide may continue to present it intermittently without the apparatus **100**, to ensure that the mastery is maintained over time.

(113) The apparatus **100** can also be used for teaching tasks related to hygiene. Tasks related to hygiene may include putting on deodorant, brushing teeth, washing hands, brushing hair, washing hair/shower and blow-drying hair. With the exception of washing hair, these tasks should be performed in front of a mirror to aid learning.

(114) During the beginning phase, the task is introduced. The object (e.g., deodorant, shampoo, etc.) in held in the student's non-dominant hand, and the student's dominant hand is used to twist off, pull off, or push off a lid/cap. The guide may use his/her own strength at first and move slowly. The guide can use declarative statements such as: ‘this cap is hard to twist off’. The guide can also use vocal sounds instead of words, such as a grunt, to convey difficulty of twisting off a cap. In another example, the toothbrush is held in the dominant hand, while the toothpaste is squeezed with the non-dominant hand. Similarly, the comb/hairbrush/blow-dryer is held in the dominant hand.

(115) The guide can quantify how much deodorant, toothpaste, etc. is used, by counting ‘1,2,3’ or whatever the guide thinks may serve such purpose. The guide may use his/her own strength, moving very slowly and if needed, exaggeratingly slow, to rub/spray deodorant, and then switch the deodorant to other hand and repeat. The guide can use declarative statements such as: ‘putting on deodorant to smell nicely’, ‘we need deodorant because we sweat’, ‘shampoo makes our hair clean’, ‘clean teeth are happy teeth’, ‘let's squeeze out the toothpaste’, etc. The guide can even sing a song, ‘this is the way we brush our teeth, brush our teeth, brush our teeth’, etc.

(116) During the middle phase, there may be an increase in the student's motivation to act. The beginning phase may blend into the middle phase with the exception that during the middle phase, the guide is consciously decreasing his own strength in the task, and allowing for the student to take over.

(117) With deodorant in the dominant hand, the guide may just wait for student to raise it to his/her armpit. If the student does not yet move, then the guide can apply a little pressure, as a prompt for the student to raise the deodorant to his/her armpit. Once at the armpit, the student may be allowed to use his/her own strength in applying the deodorant. If student does not move his hand, then the guide may aid by just moving it a little bit, serving as a prompt for the student to continue. The guide can then prompt ‘1,2,3’, counting as slowly as needed. After the deodorant is applied, it is important that the student completes the task by placing the cap back on the deodorant container. A declarative statement, such as “the cap needs to go back on the deodorant”, may be used. If the student does not do so on his own, then the guide can place hand on the cap and just wait for the student to pick it up and complete task. Reinforcements may be used if required.

(118) During the end phase, the student's mastery of the task is tested. At this point, the guide believes that the student is using own strength to complete the task. The apparatus **100** may be removed and the task is presented to the student. If the student does not respond independently after some time, then the guide can add some verbal prompts, i.e., “hmmm, how do we get the toothpaste out?”. It is paramount that the guide does not have any expectations, but maintain a state of neutrality, by being patient and giving the student time to act. If the student can perform the task independently (without the tool or the apparatus **100**), then he/she has mastered the task. The guide should continue to present the task (without the tool or the apparatus **100**) during sessions and intermittently, as time goes on. If the student begins the task (without the tool or the apparatus **100**), but then stops, the guide can then place his/her hand over the student's hand and just wait for a response.

(119) The results may be contingent upon a frequency and consistency of use/practice. As in any

other type of learning or therapy, the more it is applied, the greater the possibilities of learning, and at a faster rate.
(120) The length of a session should be specific to the student's capabilities. If a session proves to be too long or overwhelming for the student, then the student may become disinterested and/or disengaged. The guide must be mindful of the student's mental status at all times. It may be appropriate to begin with just a few minutes and add to a session time, as time progresses. If time allows, the guide may also attempt multiple times during a given day, at different times of the day. A recommended use is every day, during the beginning of a task. If that is not possible, then the recommended minimum use is every alternative day.
(121) The apparatus **100** can also be used for teaching tasks related to eating. The tasks related to eating include the ability to hold a fork or spoon and feed oneself, cutting food with a knife and fork, and pouring a drink and so forth.
(122) During the beginning phase, the task is introduced to the student by using the apparatus **100**. In an example, the spoon/fork is held in the student's dominant hand for spooning/piercing food. The guide may use his/her own strength at first and moves slowly taking food and bringing it to the student's mouth. The guide can use declarative statements, such as “Mmm . . . this cereal/pasta/eggs tastes good”, or “Let's see how this pasta tastes”, etc. It is suggested to use food that the student likes, so as to motivate student.
(123) For the task of cutting a piece of meat or vegetable, the fork is held in the student's non-dominant hand and the knife is held in the dominant hand. Declarative statements can include “wow, this carrot is hard to cut”, “let's cut this chicken so you can eat it”, “we need to be very careful with the knife and cut very slowly”, etc.
(124) For the task of pouring a drink, a container that is not too heavy with liquid (e.g., milk/juice/water) may be used. However, the container may be heavy enough so that it provides a challenge for the student. Further, plastic cups/mugs may be used during the beginning phase.
(125) During the middle phase, the guide may begin to relinquish strength, while allowing the student to apply his/her own strength in task. Declarative language, such as “I need help cutting this potato/pouring this milk”, etc. may be used. The guide may wait for the student to act. If the student does not move, then the guide can give a gentle tug to guide the student. It may be important that the guide is continually assessing the student's ability in completing the task. Reinforcements may be used whenever required.
(126) During the end phase, the guide may believe that the student is using his/her own strength to complete the task. The apparatus **100** may be removed and the task may be presented to the student. If the student does not respond independently after some time, then the guide can add some verbal prompts, i.e., “Let's cut this cucumber”, “Time to pour some juice”, etc. It is paramount for the guide to not have any expectations, but maintain a state of neutrality. It is also important for the guide to be patient and give the student time to act.
(127) If the student can perform the task independently (without the tool or the apparatus **100**), then he/she has mastered the task. The guide may continue to present the task (without the tool or the apparatus **100**) during sessions and intermittently, as time goes on.
(128) If the student begins the task (without the tool or the apparatus **100**), but then stops, the guide can then place his hand over student's hand and just wait for a response.
(129) Results may be contingent upon a frequency and consistency of use/practice. As in any other types of learning or therapy, the more it is applied, the greater the possibilities of learning, and at a faster rate.
(130) The apparatus **100** can also be used for teaching tasks related fastening clothes. The tasks related to fastening clothes may include self-dressing activities, such as fastening buttons, snaps, zippers and tying shoes and so forth.
(131) During the beginning phase, the task is introduced to the student. It is suggested to begin teaching this task through the use of oversized buttons, zippers, snaps and shoe laces, as can be

found on dolls or in therapeutic tools/toys. The guide may use his/her own strength at first and move slowly. As the student masters each of these, then the size of buttons and snaps can be reduced over time. Declarative language, such as “let's get this buttoned”, “this is how we tie a shoe”, “this snap is tricky”, etc., may also be used.

(132) During the middle phase, the student's motivation to act is increased. The guide may begin to relinquish strength, while allowing the student to apply his/her own strength in task. The guide may wait for the student to act. If the student does not move, then the guide can give a gentle tug to guide the student. It is important that the guide continually assesses the student's ability in task, and therefore use declarative language accordingly.

(133) For tying shoelaces, it is suggested to practice on shoes while they are not being worn. The guide may start the task as one would typically tie shoes, and place the shoes in front of the student, with the heel of each shoe closest to the student. The student may use the dominant hand in tying, while holding the first looped lace in place with the non-dominant hand.

(134) For buttoning, the size of the button should become smaller and smaller, until the student is able to button clothing (e.g., a shirt, trousers, etc.). The student may use the dominant hand to push button through button hole. Reinforcements may be used whenever required.

(135) During the end phase, the guide believes that the student is using his/her own strength to complete the task. The apparatus **100** is removed and the task is presented to the student. If the student does not respond independently after some time, then the guide can add some verbal prompts, i.e., “Let's tie these shoes”, “This button has to go through this slit”, etc. It is paramount that the guide does not have any expectations, but maintain a state of neutrality, by being patient and giving student time to act.

(136) If student can perform the task independently (without the apparatus **100**), then he/she has mastered the task. The guide may continue to present the task (without the apparatus **100**) during sessions and intermittently, as time goes on. If the student begins the task (without the apparatus **100**), but then stops, the guide can then place his hand over the student's hand and just wait for a response.

(137) Results may be contingent upon frequency and consistency of use/practice. As in any other type of learning or therapy, the more it is applied, the greater the possibilities of learning, and at a faster rate.

(138) The apparatus **100**, **200**, **300**, **400**, **500**, **600**, **700**, or **800** can be a teaching or training tool for a classroom, a therapist's clinic or a home. The apparatus **100**, **200**, **300**, **400**, **500**, **600**, **700**, or **800** can be used by parents of learning disabled and autistic children, special needs educators and institutions, and/or occupational therapists and institutions.

(139) FIGS. **9**A-**9**F illustrate various views of an apparatus **900** suitable for use by a younger or smaller patient, e.g., one who is within an age range of about 3-9 years old. FIG. **9**A is top part view, FIG. **9**B is a bottom part view, and FIG. **9**C is a side view depicting an attachment of apparatus **900**. FIG. **9**D is a bottom part view of fingers, band, and attachment on a thumb side. FIG. **9**E illustrates details of the outside seams, and FIG. **9**F shows a top part view of the fingers portion of apparatus **900**. Apparatus **900** may further include a band on the child side, either as a replacement for or in addition to a child-sized glove. The band would run transversely across a child's palm when the child's hand is inserted into apparatus **900**.

(140) FIGS. **9**G-**9**I illustrate various views of usage of apparatus **900**. Apparatus **900** may be used in at least two ways, e.g., utilizing the band for a toddler, or utilizing the bottom part for a child with a bigger hand. FIG. **9**G is a top view, and FIG. **9**H is a bottom view, of apparatus **900** during usage. FIG. **9**I illustrates a typical usage scenario of apparatus **900**. Apparatus **900** differs from apparatus **100** at least by having shaped the adult side (also referred to as a parent side, guide side, teacher side, etc.) being shaped to facilitate entry and exit of hands from apparatus **900**.

(141) Furthermore, apparatus **900** differs from apparatus **800** by having deleted the elastic bands from the fingers, and the palm elastic **832** is modified into a wider band on the child side of

apparatus **900**. The wider band may accommodate a hand of a toddler or small child to slip into apparatus **900** with ease. The adult (or parent, guide, teacher, etc.) can still have control, while the child can still feel movement and amount of pressure from above during demonstration and action.

(142) In particular, apparatus **900** may include two gloves or apparatuses sized accordingly, one glove for an adult and the other glove for a child, connected for hand-over-hand tactile and sensory demonstration and action. The gloves may be attached at their sides in order to provide improved control and feel (e.g., tactile feedback) during activity demonstration and action. Apparatus **900** may include minimal attachment in order to allow greater movement and flexibility while maintaining control. Apparatus **900** may be free (i.e., not coupled together) at the wrists in order to allow greater range of motion. Apparatus **900** may be fingerless in order to allow for improved feel (e.g., tactile feedback) and to strengthen dexterity.

(143) Apparatus **900** may have an adult side that is shaped for easy entry and exit but still having enough coverage (e.g., covering the knuckles and halfway down back of hand) for control and feel during activity demonstration and action. The shape allows for greater ease for an adult when putting on apparatus **900**, as well as an ability to quickly disengage the adult hand from apparatus **900** if necessary.

(144) Apparatus **900** may include a child side sized for smaller hands of younger ages (approximate ages 3 to 9 depending upon child), with an additional feature of an attached band at the near top (palm side) in order to allow smaller hands, including toddler-size hands. A smaller and/or toddler size hand would simply slide into the band, and the adult would control motion and demonstration from their part of apparatus **900** while observing from above the child.

(145) Apparatus **900** may be sewn, heat-sealed, fastened with hook and loop fasteners like Velcro®, or attached with clips, pins, snaps, and so forth (generically, "attachment features"), in order to hold together the top part to the bottom part. Apparatus **900** may include an additional feature of an attached band at the near top (palm side) to allow smaller and/or toddler size hand. The band may be secured in place by an attachment feature as previously discussed.

(146) Apparatus **900** may be constructed from a stretchable fabric, a compression or compressible fabric, and a material with additional properties such as being quick dry, stain-resistant and easily washable, and so forth. Compared to apparatus **100**, apparatus **900** may provide an adult side that is shaped to facilitate entry and exit.

(147) Apparatus **900** improves upon apparatus **800** by deleting the material bands from the fingers, and having palm material band **832** be modified into a wider band on the child side of the apparatus. The wider band is to accommodate the hand of a toddler or small child slipping in with ease, and whereby the Adult, Parent, Guide and or Teacher can still have control, and the child/toddler can still feel movement and amount of pressure from above during demonstration and action.

(148) Apparatus **900** may include a further improvement of having all seams located on an outside surface of at least the bottom part in order to provide a smooth, seamless finish and tactile sensation in the interior of the apparatus to accommodate sensitivity and sensory aversion. In some embodiments, both the top part and bottom part may have seams located on outer surfaces of the respective parts. Such embodiments may have no seams on inner surfaces, i.e., no seams on surfaces facing an inner cavity of the part which would be in direct contact with a hand during normal usage. Apparatus **900** may further accommodate toddlers and small-hand children beyond the use of the attached band, because as a child grows and their hands get larger, the child may use the glove part and continue learning utilizing apparatus **900**.

(149) FIG. **10** illustrates an apparatus **1000** suitable for use by an older child, grown adult patient, or other patient with a relatively large hand, e.g., one who is at least 10 years old. FIG. **10**A illustrates a top part view, usable by an adult. FIG. **10**B illustrates a top part view of a fingers portion and the outside seam detail. FIG. **10**C illustrates a top part view of an adult side of apparatus **1000**. FIG. **10**D illustrates a bottom part view of a child side of apparatus **1000**. FIG. **10**E

illustrates a side view depicting an attachment at a thumb and index finger location. FIG. **10**F illustrates a side view depicting an attachment at a fourth (i.e., little finger) side. FIG. **10**G illustrates a usage application, specifically learning to brush teeth.

(150) Apparatus **1000** differs from apparatus **100** at least by having the adult side (or parent, guide, teacher side, etc.) of apparatus **1000** being shaped to facilitate entry and exit of the adult hand.

(151) Apparatus **1000** top and bottom portions may be attached at their sides in order to provide improved for control and tactile feel and feedback during an activity demonstration and action.

(152) Apparatus **1000** minimizes attachments in order to allow greater movement and flexibility while maintaining control. For example, apparatus **1000** may be free at the wrists in order to allow greater range of motion, and apparatus **1000** may be fingerless in order to allow improved feel and strengthen dexterity.

(153) Although the adult side of apparatus **1000** may be shaped to provide easier entry and exit, apparatus **1000** still has enough coverage (e.g., at least covering the knuckles and half-way down the back of the hand) to provide improved control and feel during an activity demonstration and action. This feature allows greater ease for the adult when putting on the apparatus **1000**, as well providing an ability to quickly disengage if necessary.

(154) Apparatus **1000** may include a child side that is sized for larger hands of older children, teenagers, or another adult-sized person (e.g., older than about 10 years old).

(155) Apparatus **1000** may have all seams be located on an outside (exterior) surface of both the top part and the bottom part in order to provide a smooth, seamless finish in the interior of the apparatus to accommodate sensitivity and sensory aversion of hands within apparatus **1000**.

(156) Apparatus **1000** may be constructed from a technical fabric. A technical fabric is known as a material that provides a particular function and/or benefit (e.g., a neoprene, an elastane such as Lycra® or Spandex®, a softshell jersey, etc.). A technical fabric may be used with apparatus **1000** to provide desired levels of stretch, compression, and additional properties such as quick dry, stain-resistant and easy washability and so forth.

(157) FIG. **11** illustrates a method **1100** of using embodiments of the invention for training or educating a student through tactile sensory demonstration and action. The method **1100** may be implemented using any of the apparatus **100**, **200**, **300**, **400**, **500**, **600**, **700**, **800**, **900**, or **1000**. For descriptive purposes, the method **1100** will be described with reference to the apparatus **100** (shown in FIGS. **1**A-**1**C).

(158) At step **1102**, the first part **102** of the apparatus **100** is worn by a guide. Specifically, the guide may insert his/her hand through the wrist opening **106***a* so that the hand is enclosed by the body **103** and the fingers are partially received within the corresponding finger elements **108**, **110**, **112**, **114** and **116**. In an embodiment, the guide may wear the first part **102** on each of the hands.

(159) At step **1104**, the second part **104** of the apparatus **100** is worn by a student. Specifically, one hand of the student may be inserted through the wrist opening **122***a* so that the hand is enclosed by the body **105** and the fingers are partially received within the corresponding finger elements **124**, **126**, **128**, **130** and **132**. In an embodiment, the student may wear the second part **104** on each of the hands.

(160) At step **1106**, the guide may control movements of the student and apply pressure, as required, using the first part **102** of the apparatus. Therefore, the first part **102**, worn by the guide, is used to control the movements of the second part **104** worn by the student. In case, the apparatus **100** is worn on two hands, the guide may control both hands of the student. The controlled movements and applied pressure may depend upon a task that is being taught to the student. The task may be any task, for example, but not limited to, opening a bottle, performing a hygiene related task, eating, fastening clothes/shoes, manipulating scissors, writing and so forth.

(161) Though the apparatus **100**, **200**, **300**, **400**, **500**, **600**, **700**, **800**, **900**, and **1000** as described above, include one or more gloves, the present disclosure may also include a hand-over-hand apparatus having any first part configured to be secured to a hand of a guide, and any second part

configured to be secured to a hand of a student. The first part and/or the second part may not be a glove.
(162) Further, in an embodiment, the apparatus **100**, **200**, **300**, **400**, **500**, **600**, **700**, **800**, **900**, and **1000** may be made of a similar material.
(163) While the foregoing is directed to embodiments of the present disclosure, other and further embodiments of the present disclosure may be devised without departing from the basic scope thereof. It is understood that various embodiments described herein may be utilized in combination with any other embodiment described, without departing from the scope contained herein. Further, the foregoing description is not intended to be exhaustive or to limit the disclosure to the precise form disclosed. Modifications and variations are possible in light of the above teachings or may be acquired from practice of the disclosure. Certain exemplary embodiments may be identified by use of an open-ended list that includes wording to indicate that the list items are representative of the embodiments and that the list is not intended to represent a closed list exclusive of further embodiments. Such wording may include “e.g.,” “etc.,” “such as,” “for example,” “and so forth,” “and the like,” etc., and other wording as will be apparent from the surrounding context.
(164) No element, act, or instruction used in the description of the present application should be construed as critical or essential to the disclosure unless explicitly described as such. Also, as used herein, the article “a” is intended to include one or more items. Where only one item is intended, the term “one” or similar language is used. Further, the terms “any of” followed by a listing of a plurality of items and/or a plurality of categories of items, as used herein, are intended to include “any of,” “any combination of,” “any multiple of,” and/or “any combination of multiples of” the items and/or the categories of items, individually or in conjunction with other items and/or other categories of items.
(165) Moreover, the claims should not be read as limited to the described order or elements unless stated to that effect. In addition, use of the term “means” in any claim is intended to invoke 35 U.S.C. § 112(f), and any claim without the word “means” is not so intended.

Claims

1. An apparatus for learning by physical demonstration, comprising: a first glove adapted to a human hand of a guiding instructor, the first glove comprising: a palm-side outer surface; a dorsal-side outer surface opposite from the palm-side outer surface; and one or more finger portions adapted to at least partially enclose a respective finger; and a second glove sized and shaped to accommodate a human hand of a student, wherein a size of the first glove is different than a size of the second glove, the second glove comprising: a palm-side outer surface; a dorsal-side outer surface opposite from the palm-side outer surface; and one or more finger portions adapted to at least partially enclose a respective finger, wherein the second glove is configured to removably couple in a palm-to-dorsal orientation with the first glove to permit guided movement of the human hand of the student by the human hand of the guiding instructor such that the human hand of the student mimics movements made by the human hand of the guiding instructor during performance of a manual task, and wherein seams of the first glove or second glove are located only on an outer surface of the first glove or second glove, respectively.
2. The apparatus of claim 1, wherein the second glove is smaller than the first glove.
3. The apparatus of claim 1, further comprising a finger attachment between a finger portion of the first glove and a finger portion of the second glove.
4. The apparatus of claim 3, wherein the finger attachment comprises a material band.
5. The apparatus of claim 4, wherein the material band comprises a material selected from a group consisting of an elastic material and a fabric material.
6. The apparatus of claim 1, wherein a finger portion of the first glove and a corresponding finger portion of the second glove are open-ended.

7. The apparatus of claim 1, wherein the second glove comprises a material selected to accommodate sensory sensitivities associated with autism spectrum disorder, the material comprising at least one of a natural fabric, synthetic fabric, wool, leather, rubber, latex, and neoprene.
8. The apparatus of claim 1, wherein the dorsal-side of the second glove is coupled to the palm-side of the first glove.
9. The apparatus of claim 1, wherein the second glove is coupled to the first glove only at one or more finger sections.
10. The apparatus of claim 1, wherein a wrist portion of the second glove remains free from a wrist portion of the first glove.
11. The apparatus of claim 1, wherein the first glove is larger in size than the second glove.
12. The apparatus of claim 11, wherein the first glove is configured to fit a right human hand or a left human hand of the guiding instructor, and wherein the second glove is configured to fit a right human hand of the student or a left human hand of the student corresponding to a directive hand of the guiding instructor, the apparatus further comprises: a third glove configured to fit the other of a left or right human hand of the guiding instructor, the third glove comprising: a palm-side outer surface; a dorsal-side outer surface opposite from the palm-side outer surface; and one or more finger portions adapted to at least partially enclose a respective finger, and a fourth glove configured to fit the other of a left or right human hand of the student, the fourth glove comprising: a palm-side outer surface; a dorsal-side outer surface opposite from the palm-side outer surface; and one or more finger portions adapted to at least partially enclose a respective finger.